18. THE VENOUS SYSTEM : NOTES

DURATION : 07:29 STUDY SHEET

COURSE SUMMARY

This video delves into the crucial role of the venous system, addressing topics such as the vertebral plexuses and their connection to the nervous system, as well as the importance of effective venous drainage for bodily health. You will learn to identify and release venous compressions that can lead to chronic pain, such as neck pain or pelvic disorders. Furthermore, it explains how abdominal pressure and diaphragmatic function impact circulatory balance, with direct implications for emotional and physical well-being. Practical techniques for improving lymphatic and venous drainage will also be described, allowing you to apply this knowledge in your therapeutic practice.

THE VENOUS SYSTEM

THE VERTEBRAL PLEXUSES AND THE NERVOUS SYSTEM

At a neurological level, the **vertebral plexuses** are composed of several essential parts:

- The **deep medullary system**.
- The **internal vertebral venous system**, located between the dura mater and the dural sinus.
- The **external vertebral venous plexus**.
- The **diploic systems** and the **emissary veins**, more specific to the skull.

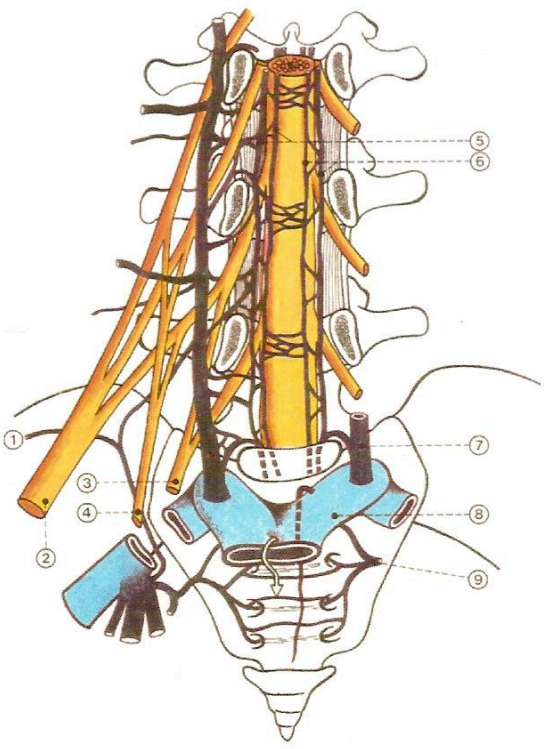


Figure 12. Rapports des plexus vertébraux. Référence (36), p 116.

FIG. — Vertebral plexuses

There is also a **medullary plexus** and an **internal plexus**, notably the **pterygoid sinus**, which can be a site of congestion. It is crucial to learn how to release this system, which acts as a safety valve in case of engorgement.

In cases of **chronic cervicalgia** or **acute torticollis**, these are often venous compressions. Inside a nerve (perineurium, endoneurium, epineurium), small vessels can become engorged with blood and compress the neuronal system. Our role is then to **decongest** these areas.

Discopathy occurs when **venous stasis** releases **cytokines** and creates acidity that destroys the system. Rebalancing pressures becomes paramount. The **dural sinus** is continuous with the vertebral plexus, and the mobility of the **sacrum** can influence headaches.

VENOUS DRAINAGE AND CRANIAL SINUSES

Venous blood flows towards the **cranial sinuses**: the **straight sinus**, the **superior sagittal sinus**, the **inferior sagittal sinus**, the **straight sinus**, and the **transverse sinus**.

These systems are connected to the **fourth ventricle**, particularly the **foramen of Magendie**, which is an opening involved in the pulsation of the **cerebrospinal fluid (CSF)** system. By releasing the vascular system from the periphery to the deep, we reopen the vascular fields and

redrain the entire system.

This drainage occurs from the **carotids**, the **vertebral arteries**, and all the sinuses via the vertebral plexus, directly involving a free sacrum.

THE FUNCTIONAL UNIT OF PELVIC VEINS AND LOWER LIMBS

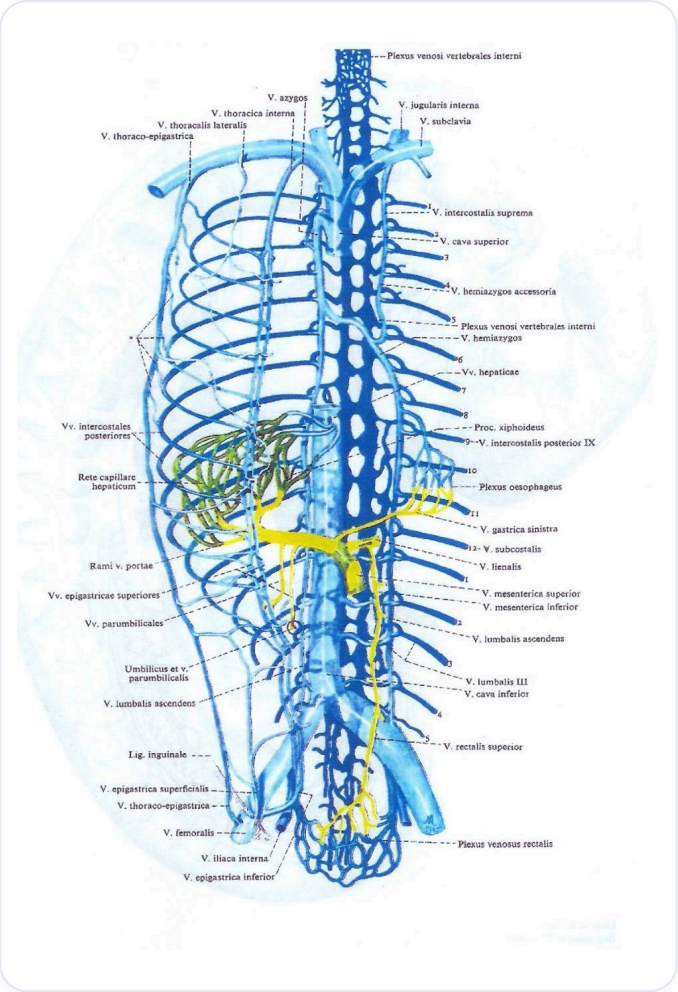


FIG. — Human venous system

The **pelvic veins** and those of the **lower limbs** constitute a functional unit. An imbalance in the lower limbs (feet, knees, pelvis) has repercussions on the **urogenital system**.

For example, walking activates drainage thanks to **Lejars' sole** and the **navicular-cuboid articulation**, which functions like a diaphragm. In cases of small pelvis disorders in women, it is essential to check and release the foot to restore this return force, otherwise stasis may occur.

CONGESTIONS AND COLLATERAL PATHWAYS



Figure 11. ...parmi beaucoup d'autres: les variances simples, doubles et plexiformes de la veine iliaque interne. Référence (37).

FIG. — Internal iliac vein

Lower limb pain can be related to **nerve compression** that is not of urogenital origin. Certain **circulatory collateral pathways** are the reappearance of a foetal state that regresses under physiological conditions.

Uterine retroversions can lead to frequent congestions. The paucity of **venous valves** promotes bidirectional circulation, allowing blood to change direction in case of congestion, but can also create areas of local stasis (prostate, etc.).

Abdominal pressure influences the path of venous blood. An increase in this pressure can direct blood towards the vertebral plexuses, leading to vertebral congestions and **low back pain** of digestive origin. It is often necessary to restore pressures to relieve the liver and restore dynamic balance.

IMPACT OF VENOUS CIRCULATORY DISORDERS

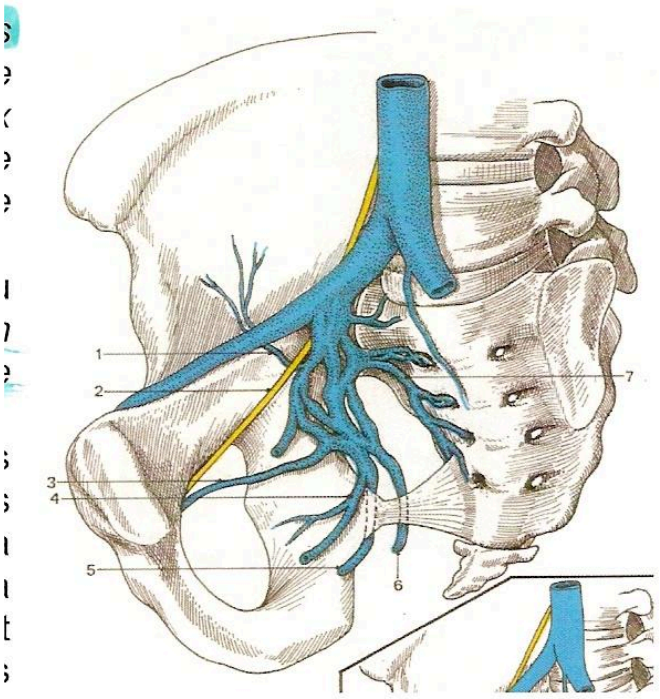


Figure 10. La veine iliaque interne: une disposition possible, mais ... Référence (36,) p 108

FIG. — Posterior view pelvic vascularization

Disorders of venous circulation can be the cause of multiple symptoms:

- Persistent deep **dyspareunia**.
- **Dysmenorrhoea**.
- **Constipation**.
- **Tenesmus**.
- General impairment (asthenia, depressive tendencies, emotional irritability).

THE DIAPHRAGMATIC PUMP AND LYMPHATIC DRAINAGE

The great key lies in the diaphragmatic pump, which acts on the **vascular**, **arterial**, and **lymphatic** systems simultaneously. The development of the vascular system generates blood pressure in the capillaries, creating exudates in the intercellular network. This network forms new pathways, particularly longitudinal ones.

The **cisterna chyli** (or Pécquet's cistern) is a very important network, responsible for a large amount of resorption at the level of the left supraclavicular fossa. The **psoas muscles**, thanks to the **perirenal fascia**, drain up to 1.5 litres of lymph per day through their movements.

The psoas is a thick organ that can retain toxins. **Toxemic overload** can block it, preventing the release of toxins from the peritoneum into the venous and lymphatic systems. A small muscle in the neck, the **levator scapulae muscle**, has a similar structure to the psoas and also participates in the uptake of toxicity.

THE SEVEN DIAPHRAGMS

We consider that there are **seven crucial diaphragms** for the body:

1. The feet

2. The knees

3. The small pelvis

4. The thoraco-abdominal diaphragm

5. The cerebellum and its balance with the base of the skull

6. The parietals

7. The most important drainage space for the CSF system, above the parietals.

The parietals, like **solar panels**, must breathe and capture light, just as the **radii** are "solar bones". The integration of these systems into the brain, in connection with the **vitelline** and **umbilical** systems and the **apical impulse**, allows the brain to relax.

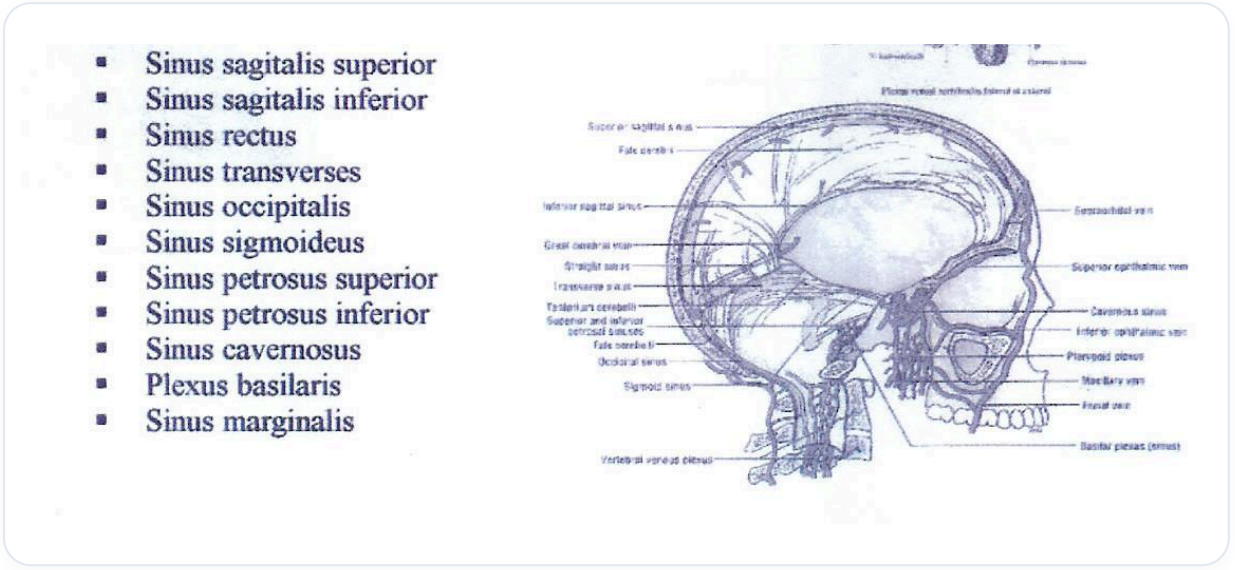


FIG. — Cranial venous sinuses